

OpenCap Body Scanning Guide

Overview

Final Evolution Lab integrates [OpenCap](https://www.opencap.ai/) (https://www.opencap.ai/) — an open-source markerless motion capture system — to provide 3D biomechanical body scanning using only smartphone cameras.

How It Works

Recording Setup

1. **Set up 2 cameras** (front and side view, ~45° apart)
2. **Ensure good lighting** — avoid backlighting
3. **Wear form-fitting clothes** — avoid baggy clothing
4. **Clear the area** — 3m × 3m minimum space
5. **Camera height** — waist level, 3-4 meters away

Recording Process

1. **Calibration** — Perform T-pose and basic movements
2. **Movement Sequence** — 30-60 second recording
3. **Upload** — Videos uploaded to OpenCap server
4. **Processing** — AI generates 3D skeleton (2-5 minutes)
5. **Results** — Full biomechanical analysis delivered

Recommended Movement Sequences

Scan Type	Movements	Duration
Full Body	Walk, squat, jump, reach	60s
Upper Body	Arm circles, overhead press, rotation	30s
Lower Body	Squat, lunge, single-leg balance	30s
Basketball	Shooting form, defensive slide, jump	45s

Analysis Features

Joint Analysis

- **Range of Motion (ROM)** — All major joints graded Excellent→Poor
- **Joint Angles** — Measured during movements
- **Asymmetries** — Left vs. right comparison
- **Mobility Limitations** — Identified and flagged

- **Flexibility Scores** — Benchmarked against population

Movement Patterns

- **Squat Depth & Form** — Knee tracking, hip hinge, torso angle
- **Jump Mechanics** — Triple extension, arm swing, landing
- **Running Gait** — Stride length, cadence, foot strike
- **Shooting Form** — Elbow alignment, release point

Performance Metrics

- Vertical Jump Height (cm)
- Sprint Speed (m/s)
- Agility Score (0-100)
- Power Output (Watts)
- Endurance Capacity (0-100)
- Flexibility Score (0-100)
- Balance Score (0-100)

Form Correction AI

Real-time Feedback Methods

1. **Visual Overlay** — Skeleton comparison on screen
2. **Audio Cues** — “Lower your hips”, “Straighten your back”
3. **Haptic Feedback** — Controller vibration patterns
4. **Text Instructions** — On-screen correction messages
5. **Video Demonstrations** — Reference clips

Sensitivity Levels

- **Beginner** (0.3) — Lenient, major issues only
- **Intermediate** (0.5) — Balanced feedback
- **Advanced** (0.7) — Strict form requirements
- **Elite** (0.9) — Clinical-grade precision

Injury Prevention

Risk Assessment

- Movement compensations detected
- Imbalance identification
- High-risk pattern flagging
- Injury probability estimation
- Fatigue level monitoring

Alerts

-  **Moderate Risk** — Modify workout intensity
-  **High Risk** — Focus on mobility/corrective work
-  **Critical** — Rest recommended, seek professional help
-  **Fatigue Threshold** — Stop and recover

-  **Overtraining** — Schedule rest day within 48h

Personalized Workouts

- Based on your scan data, the AI generates:
- **Exercise selection** tailored to your biomechanics
 - **Set/rep schemes** for your fitness level
 - **Progression rate** based on recovery capacity
 - **Alternative exercises** for any limitations
 - **Periodization** aligned with your goals

Training Goals

- Increase Vertical Jump
- Improve Speed
- Enhance Agility
- Build Strength
- Improve Flexibility
- Build Endurance
- Injury Rehabilitation
- General Fitness

UE5 Integration

C++ Classes

File	Purpose
FELOpenCapIntegration.h/.cpp	OpenCap API connection, video upload, session management
FELBiomechanicsAnalyzer.h/.cpp	Joint ROM analysis, movement pattern detection
FELFormCorrectionAI.h/.cpp	Real-time form correction during exercises
FELInjuryPrevention.h/.cpp	Risk assessment, fatigue tracking, alerts
FELPersonalizedWorkouts.h/.cpp	AI workout generation, training plans

Blueprint Events

- `OnBodyScanComplete` — Scan finished with results
- `OnFormCorrectionIssued` — Real-time correction available
- `OnInjuryAlertTriggered` — Risk alert fired
- `OnWorkoutGenerated` — New workout ready

API Reference

OpenCap API Endpoints

POST /v1/sessions	☐ Create new session
POST /v1/sessions/{id}/upload	☐ Upload video files
POST /v1/sessions/{id}/calibrate	☐ Run calibration
GET /v1/sessions/{id}/status	☐ Check processing status
GET /v1/sessions/{id}/results	☐ Get skeleton data
GET /v1/health	☐ Connection test

Privacy & Data

- Videos processed server-side, deleted after analysis
- Skeleton data stored in user's encrypted profile
- Body metrics never shared with third parties
- Users can delete scan history at any time
- HIPAA-compliant data handling